

1 Solution — GIAM 3.4

Problem 6

1.1 Problem

True or false: there are two irrational numbers whose sum is rational. Prove your answer.

1.2 Answer

True. $\sqrt{2}$ and $-\sqrt{2}$ are irrational and $\sqrt{2} + (-\sqrt{2}) = 0$.

1.3 Proof

The claim is **existential**,

$$\exists x, y \in \mathbb{R} \setminus \mathbb{Q}, \quad x + y \in \mathbb{Q},$$

so it is established by exhibiting one pair and verifying that it qualifies.

Proof. Let $x = \sqrt{2}$ and $y = -\sqrt{2}$.

x is irrational — this is Theorem 1.6 of the text.

y is irrational: suppose instead $-\sqrt{2} = q$ for some $q \in \mathbb{Q}$. Since $\mathbb{Q}$ is closed under negation, $-q$ is rational, and $\sqrt{2} = -q$, contradicting the irrationality of $\sqrt{2}$.

Finally,

$$x + y = \sqrt{2} + (-\sqrt{2}) = 0,$$

and $0 = 0/1$ is rational.

So x and y are two irrational numbers whose sum is rational. ■

1.4 The Picture

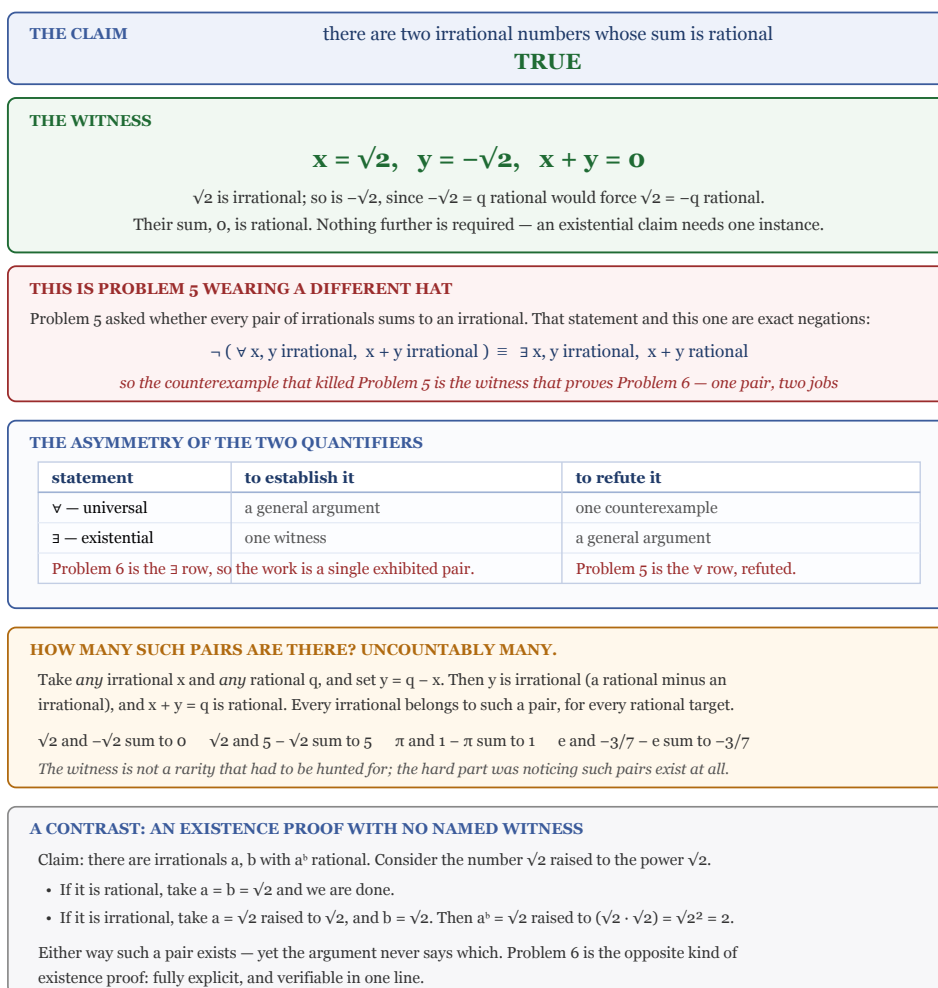


Figure 1.1: A diagram in five bands. The first states the claim — there are two irrational numbers whose sum is rational — and the verdict, TRUE. The second gives the witness: x equals root 2, y equals minus root 2, and x plus y equals 0. Root 2 is irrational, and so is minus root 2, since minus root 2 equal to a rational q would force root 2 to equal minus q , also rational. Their sum, 0, is rational, and nothing further is required because an existential claim needs one instance. The third band observes that this is Problem 5 wearing a different hat: Problem 5 asked whether every pair of irrationals sums to an irrational, and that statement and this one are exact negations — the negation of for all x and y irrational, x plus y is irrational, is precisely there exist x and y irrational with x plus y rational. So the counterexample that

killed Problem 5 is the witness that proves Problem 6: one pair, two jobs. The fourth band tabulates the asymmetry of the two quantifiers. A universal statement is established by a general argument and refuted by one counterexample; an existential statement is established by one witness and refuted by a general argument. Problem 6 is the existential row, so the work is a single exhibited pair, while Problem 5 is the universal row, refuted. The fifth band asks how many such pairs there are and answers: uncountably many. Take any irrational x and any rational q and set y equal to q minus x ; then y is irrational, being a rational minus an irrational, and x plus y equals q is rational. Every irrational belongs to such a pair, for every rational target: root 2 and minus root 2 sum to 0, root 2 and 5 minus root 2 sum to 5, pi and 1 minus pi sum to 1, e and minus three sevenths minus e sum to minus three sevenths. The witness is not a rarity that had to be hunted for; the hard part was noticing such pairs exist at all. A final band gives a contrast — an existence proof with no named witness. Claim: there are irrationals a and b with a to the b rational. Consider root 2 raised to the power root 2. If it is rational, take a and b both equal to root 2. If it is irrational, take a to be root 2 raised to root 2 and b to be root 2; then a to the b is root 2 raised to the product of root 2 with itself, which is root 2 squared, which is 2. Either way such a pair exists, yet the argument never says which. Problem 6 is the opposite kind of existence proof: fully explicit, and verifiable in one line.

1.5 Remark — This Is Problem 5 Wearing a Different Hat

Problems 5 and 6 of this section are the same problem asked twice. Their statements are exact negations of one another:

$$\underbrace{\neg \left(\forall x, y \in \mathbb{R} \setminus \mathbb{Q}, x + y \notin \mathbb{Q} \right)}_{\text{Problem 5's claim}} \cong \underbrace{\exists x, y \in \mathbb{R} \setminus \mathbb{Q}, x + y \in \mathbb{Q}}_{\text{Problem 6's claim}}.$$

Since exactly one of a statement and its negation is true, and Problem 5's claim is false, Problem 6's *must* be true — before any witness is produced. The single pair $(\sqrt{2}, -\sqrt{2})$ therefore does two different jobs:

	Problem 5	Problem 6
claim	$\forall$ -statement	$\exists$ -statement
verdict	false	true
role of $(\sqrt{2}, -\sqrt{2})$	counterexample	witness
what it does	destroys the claim	establishes the claim

The mathematical content is identical; only the framing changes. Pairing the two exercises is a deliberate lesson: **a counterexample to a universal statement is a witness for its existential negation.** They are the same object described from opposite sides.

1.6 Remark — The Quantifier Asymmetry

The pairing also makes visible how differently the two quantifiers behave:

	to establish	to refute
$\forall x, P(x)$	a general argument	one counterexample
$\exists x, P(x)$	one witness	a general argument

Each quantifier is cheap in one direction and expensive in the other, and the costs are mirror images. This is why “true or false, prove your answer” is a genuinely different task depending on which way the answer falls: guessing wrong about the verdict means doing entirely the wrong kind of work.

The practical corollary: when a universal claim looks false, hunt for the counterexample rather than a proof — that search is bounded, and proving a false statement is not.

1.7 Remark — Witnesses Are Abundant

The proof needs only one pair, but it is worth knowing the witness was not hard to come by. For **any** irrational x and **any** rational q , the pair

$$(x, q - x)$$

works: $q - x$ is irrational by 3.3 Problem 5 (a rational minus an irrational is irrational), and $x + (q - x) = q$ is rational. A few instances:

x	$y = q - x$	$x + y$
$\sqrt{2}$	$-\sqrt{2}$	0
$\sqrt{2}$	$5 - \sqrt{2} \approx 3.5858$	5
π	$1 - \pi \approx -2.1416$	1

So every irrational number belongs to such a pair, for every rational target — uncountably many witnesses in total. Problem 5’s final remark went further still: *every real number*, rational or not, is a sum of two irrationals.

Given that abundance, why is the claim worth stating? Because the difficulty was never in finding a witness but in noticing one exists — the intuition that irrational plus irrational stays irrational is strong enough that many people accept Problem 5 unchecked.

1.8 Remark — A Contrast: Existence Without a Witness

The proof here is **constructive** — it names $\sqrt{2}$ and $-\sqrt{2}$ and verifies them. Not every existence proof can, and the classic illustration is nearby in spirit.

***Claim.** There exist irrational numbers a, b with a^b rational.*

***Proof.** Consider $\sqrt{2}^{\sqrt{2}}$, which is either rational or irrational.*

- *If it is rational, take $a = b = \sqrt{2}$ and we are done.*
- *If it is irrational, take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. Then*

$$a^b = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2,$$

which is rational.

In either case such a pair exists. ■

The argument is airtight, yet at its conclusion one still does not know *which* pair works — it splits on a fact it never determines. (Gelfond–Schneider later settles it: $\sqrt{2}^{\sqrt{2}} \approx 1.6325$ is transcendental, so the second branch is live. The proof never needed that.)

This problem’s proof is the opposite extreme: the witness is named, the verification is two lines, no cases. Both are valid existence proofs; only one tells you where to look.